FIBRED TABULATORS

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ABSTRACT. This paper defines tabulators in strict double fibration (i.e., double fibrations of double categories with tabulators). Additionally, we give a Beck-Chevalley Condition for fibred tabulators. Moreover, we show that fibred tabulators are characterized by a preservation property.

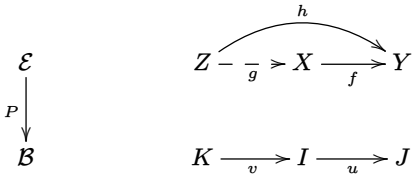
Introduction

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1 Fibrations

In this section we recall some basic facts about fibrations and their examples.

Definition 1.1. Let $P\colon \mathcal{E} \to \mathcal{B}$ be a functor. A morphism $f\colon X \to Y$ in $\mathcal{E}$ is *cartesian* over $u\colon I \to J$ i.e., $P(f) = u$ if for any $v\colon K \to I$ in $\mathcal{B}$ and $h\colon Z \to Y$ over $u \circ v$ there is a unique morphism $g\colon Z \to X$ over v with $h = f \circ g$.



A morphism $k\colon X \to Y$ is called *vertical* if it lies over an identity in $\mathcal{B}$. For an object $J \in \mathcal{B}$, the fibre of P at J denoted by $P(J)$ is the subcategory of $\mathcal{E}$ whose morphisms k lie over id_J.

Definition 1.2. A functor $P\colon \mathcal{E} \to \mathcal{B}$ is called *fibred category* or a *fibration* if for every $Y \in \mathcal{E}$ and $u\colon I \to PY$ there is a cartesian morphism $\bar{u}(E)\colon u^*E \to E$ over u. If P is a fibration, a *cleavage* for P denoted $\varphi(u, E)$ is a choice of cartesian lifts for u at E. Whence P is called a cloven fibration.

1.1 Examples

Let $\mathcal{C}^2$ denote the slice category of $\mathcal{C}$ over itself

Example 1.3 (Fundamental Fibration)**.** Let $\mathcal{C}$ be a category with pullbacks. Then the functor $\mathrm{cod}\colon \mathcal{C}^2 \to \mathcal{C}$ is a fibration; a commutative square is a cartesian arrow in $\mathcal{C}^2$ precisely when it is a pullback.

Example 1.4. The domain functor $\mathrm{dom}\colon \mathcal{C}^2 \to \mathcal{C}$ is a fibration.

Example 1.5 (Family fibration)**.** Every category $\mathcal{C}$ gives rise to a *family fibration* $\mathcal{F}\colon Fam(\mathcal{C}) \to Set$. A morphism $(u, \{\varphi_i\}_{i\in I})$ is cartesian when every φ_i is an isomorphism. Then $\mathcal{F}$ is a fibration since given $u\colon I \to J$ and $\{Y_j\}_{j\in J}$, $(u, \{1_{Y_{u(i)}}\})\colon \{Y_{u(i)}\}_{i\in I} \to \{Y_j\}_{j\in J}$ is cartesian over u.

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1.2 Fibred Finite Limits

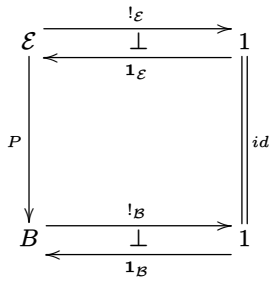
In [Str18] fibred finite limits are given in terms of terminal objects and pullbacks. Furthermore, [Jac99] captures the notion of fibred terminal objects in **Fib** using fibred adjunctions.

Definition 1.6. A fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred terminal objects if every fibre $P(I)$ has a terminal object $\mathbf{1}_{\mathcal{I}}$ and they are preserved by cartesianness.

In $\mathbf{Fib}/\mathcal{B}$, the category of fibrations over $\mathcal{B}$, a fibration P has fibred terminals if the fibred functor $F\colon P \to id_B$ has a right adjoint. Moreover, this property is characterized by preservation as follows.

Lemma 1.7. *Let $\mathcal{B}$ be a category with a terminal object $\mathbf{1}_{\mathcal{B}}$. Then $P\colon \mathcal{E} \to \mathcal{B}$ has fibred terminals if and only if $\mathcal{E}$ has a terminal $1_{\mathcal{E}}$ and $P(\mathbf{1}_{\mathcal{E}})$ is a terminal in $\mathcal{B}$.*

Proof. Consider the following diagram in **Fib**



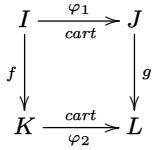
□

Futhermore, we have the following theorem

Theorem 1.8. *If $\mathcal{B}$ has a terminal object $\mathbf{1}_{\mathcal{B}}$ and $P\colon \mathcal{E} \to \mathcal{B}$ is a fibration, the following are equivalent*

(1) P has a terminal object in $\mathbf{Fib}/\mathcal{B}$.
(2) P has a terminal object in $\mathbf{Fib}$.

Lemma 1.9. *A commutative square of the form*



is always a pullback.

Definition 1.10. A fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred pullbacks if every fibre has pullbacks and they are preserved by cartesianness.

Similarly, fibred pullbacks can be characterised as follows.

Theorem 1.11. *Let $\mathcal{B}$ be a category with pullbacks. A fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred pullbacks if and only if $\mathcal{E}$ has and P preserves pullbacks.*

The following result is a consequence of Lemma 1.7 and Theorem 1.11

Theorem 1.12. *For a category $\mathcal{B}$ with finite limits, a fibration $P\colon \mathcal{E} \to \mathcal{B}$ has fibred finite limits if and only if $\mathcal{E}$ has finite limits and P preserves finite limits.*

2 Double Categories

2.1 Basic Notions

Definition 2.1. A double category $\mathbb{D}$ is a pseudo category in **Cat**, it consists of the following diagram

$$\mathbb{D}_2 \xrightarrow{\odot} \mathbb{D}_1 \underset{t}{\overset{s}{\underset{\longleftarrow}{\overset{\longrightarrow}{\longleftarrow}}}} \mathbb{D}_0$$

We call $\mathbb{D}_0$ the category of objects and vertical arrows and $\mathbb{D}_1$ the category of proarrows and cells:

$$\begin{array}{ccc} A & \xrightarrow{M} & B \\ {\scriptstyle f}\downarrow & \theta & \downarrow{\scriptstyle g} \\ C & \xrightarrow[N]{} & D \end{array}$$

We refer to M and N as the internal domain and codomain of θ respectively, whilst f and g are the external source and target respectively. Cells with proarrow (resp. vertical) identities will be called *horizontal* (resp. *vertical*) *globular* cells. Composition is strict in the vertical direction and weaker in the horizontal direction. For convention, we assume our double categories to be pseudo in the sense of [CLPS22].

Definition 2.2. Given a double category $\mathbb{D}$, we can form a *horizontal bicategory* $\mathcal{H}(\mathbb{D})$, whose objects are the objects of $\mathbb{D}$, 1-cells are the proarrows of $\mathbb{D}$, and 2-cells are given by vertical globular cells. Dually, we can form a *vertical 2-category* $\mathcal{V}(\mathbb{D})$ whose objects are the objects of $\mathbb{D}$, 1-cells are the vertical arrows, and 2-cells given by horizontal globular cells.

Notation 2.3. For cells α and β, we shall write $\frac{\alpha}{\beta}$ as their vertical composition.

2.2 Examples

Example 2.4. Let $\mathbb{D}^2$ be the slice double category of $\mathbb{D}$ over itself with external source and target operations coming from $\mathbb{D}$.

Example 2.5. Let $\mathcal{C}$ be a category with pullbacks, we define a double category $\mathbb{S}\mathbf{pan}(\mathcal{C})$ whose vertical category is $\mathcal{C}$, whose proarrows $A \nrightarrow B$ are spans $A \xleftarrow{f} C \xrightarrow{g} B$ in $\mathcal{C}$, and whose cells are commuting squares:

$$\begin{array}{ccccc} A & \longleftarrow & C & \longrightarrow & B \\ \downarrow & & \downarrow & & \downarrow \\ D & \longleftarrow & F & \longrightarrow & E \end{array}$$

in $\mathcal{C}$. Horizontal composition is given by pullbacks.

Example 2.6. The double category $\mathbb{R}\mathrm{el}(\mathrm{Set})$ has objects as sets and vertical arrows as set-functions. The proarrows are given by relations and there is a unique cell

$$\begin{array}{ccc} A & \xrightarrow{R} & B \\ {\scriptstyle f}\downarrow & \Downarrow & \downarrow{\scriptstyle g} \\ C & \xrightarrow[S]{} & D \end{array}$$

whenever $R(a,b) \implies S(f(a),g(b))$ holds for all $a \in A, b \in B$.

Definition 2.7. Let $\mathbb{E}$ and $\mathbb{B}$ be pseudo categories internal to a 2-category $\mathcal{K}$ with 2-pullbacks. A *lax functor* $L\colon \mathbb{E} \to \mathbb{B}$ consists of arrows $L_0\colon E_0 \to D_0$ and $L_1\colon E_1 \to B_1$ together with comparison morphisms given by



such that the source and target are strictly preserving. Moreover, ϕ, ι, and the associators and unitors of $\mathbb{E}$ and $\mathbb{B}$ satisfy the usual coherence conditions as in [MF06]. A lax functor is called **strict** if ϕ and ι are identities.

Remark 2.8. A strict functor in **Cat** corresponds to a strict functor between double categories.

We recall the notion of double limits i.e., limits in double categories introduced in [GP99]. However, our limits will be analogous to limits of Grandis and Paré.

Definition 2.9 ([DPP10])**.** A double category $\mathbb{E}$ has *vertical pullbacks* if the diagram

$$\mathbb{E}_2 \xrightarrow{\odot} \mathbb{E}_1 \underset{t}{\overset{s}{\underset{\longleftarrow i}{\rightrightarrows}}} \mathbb{E}_0$$

lies in **PBCat** the category of categories with pullbacks and pullback-preserving functors.

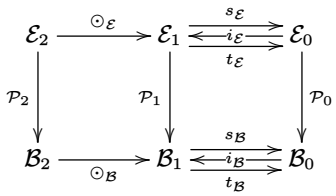
A functor $G\colon \mathbb{E} \to \mathbb{A}$ between double categories with vertical pullbacks preserves pullbacks if it preserves pullbacks componentwise. When this is the case, we shall call G a pullback-preserving functor. For the rest of the paper, we shall refer to vertical pullbacks by v-pullbacks.

Remark 2.10. The universal property of v-pullbacks in double categories generalizes the universal property of 2-pullbacks. $\odot$ preserving pullbacks amounts to a functorial choice of pullbacks, see [GP99].

3 Double Fibrations

Lemma 3.1 ([CLPS22])**.** *Given a diagram $s\colon P_1 \to P_0 \leftarrow P_1 \colon t$ in* **cFib***, a pseudo category structure in* **Fib** *(resp.* **cFib***) is the same as two pseudo double category structure obtained by applying the top and bottom 2-functors,*

Definition 3.2 ([CLPS22])**.** A strict double functor $P\colon \mathbb{E} \to \mathbb{B}$ between pseudo double categories given by



defines a **double fibration** when P_0 and P_1 are cloven fibrations, such that $s_{\mathcal{E}}$ and $t_{\mathcal{B}}$ preserve cleavages simultaneously, and $\odot_{\mathcal{E}}$ and $i_{\mathcal{E}}$ preserve cartesian morphisms.

Remark 3.3. A double fibration $P\colon \mathbb{E} \to \mathbb{B}$ is a pseudo category internal to **Fib** such that the source and target maps preserve chosen cleavages. A cleavage for P is chosen componentwise. Since $s_{\mathcal{E}}$ and $t_{\mathcal{B}}$ preserve cleavages, they are morphisms in **cFib**.

References

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